Infosys
Intelligent Bot Management

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Agenda

- Problem with typical RPA setups
- What's needed to solve the problem?
- Overview of Infosys Intelligent Bot Management Solution
    - High Level Architecture
    - Process Flow
    - Security Considerations – Access security within
- Demo
- Opensource Support – Available in Infosys GitHub
    - Hardware/software pre-requisites

Target Audience

Developers/Architects : Will understand Key Concepts and get a peek into the Technical details behind the Solution

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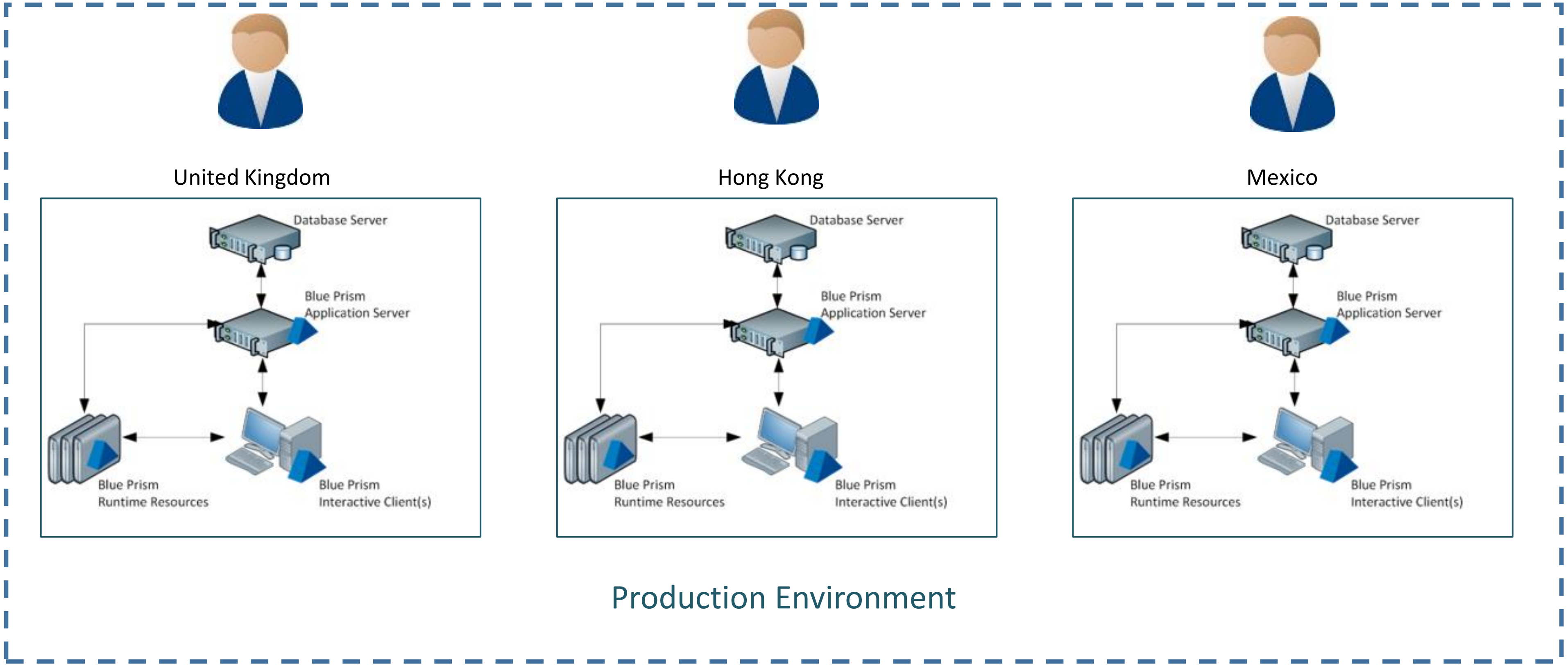
What is the Problem with typical RPA setups?

Multiple control rooms are configured in a production environment which are not easy to monitor for business owners. Control rooms are configured across multiple regions



Business Impact

- Lack of visibility
- High volume of failures
- Manual support
- BOT utilization
- Fragmented problem management

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What's needed to solve the problem?

Infosys Intelligent Bot Management for effective Outage Management & Operations Optimization

Business Manager | RPA Admin | DB Admin | IT Admin

Persona based features & Access

Monitoring
- Health Dashboards
- Anomaly Detection
- Alerts & notification

Self Heal
- Remediation plans
- Automation Engine
- Script Repository

Analytics
- Bot Executive Dashboard
- Bot Performance Dashboard
- Bot Business KPI Dashboard

Discovery
- Meta-data and Dependency Configuration

Administration
- Bot registry & onboarding
- Role and Access Management
- Components configuration
- Logs & Audit trail

Key Benefits
- Improved visibility across LoB & Bot platforms
- Minimize Business Disruptions
- Alignment between IT & Business KPIs
- Faster resolution of Bot issues

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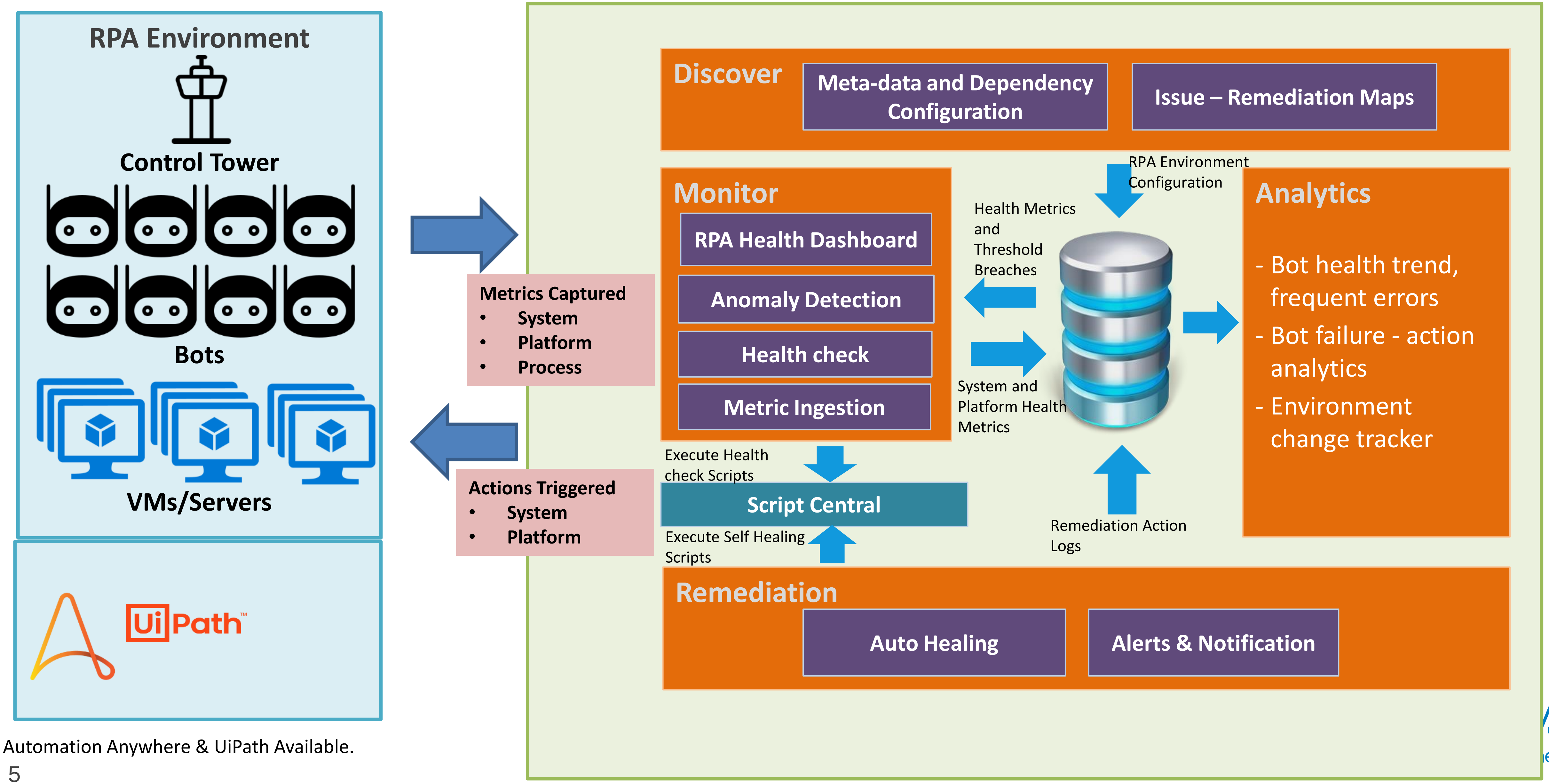
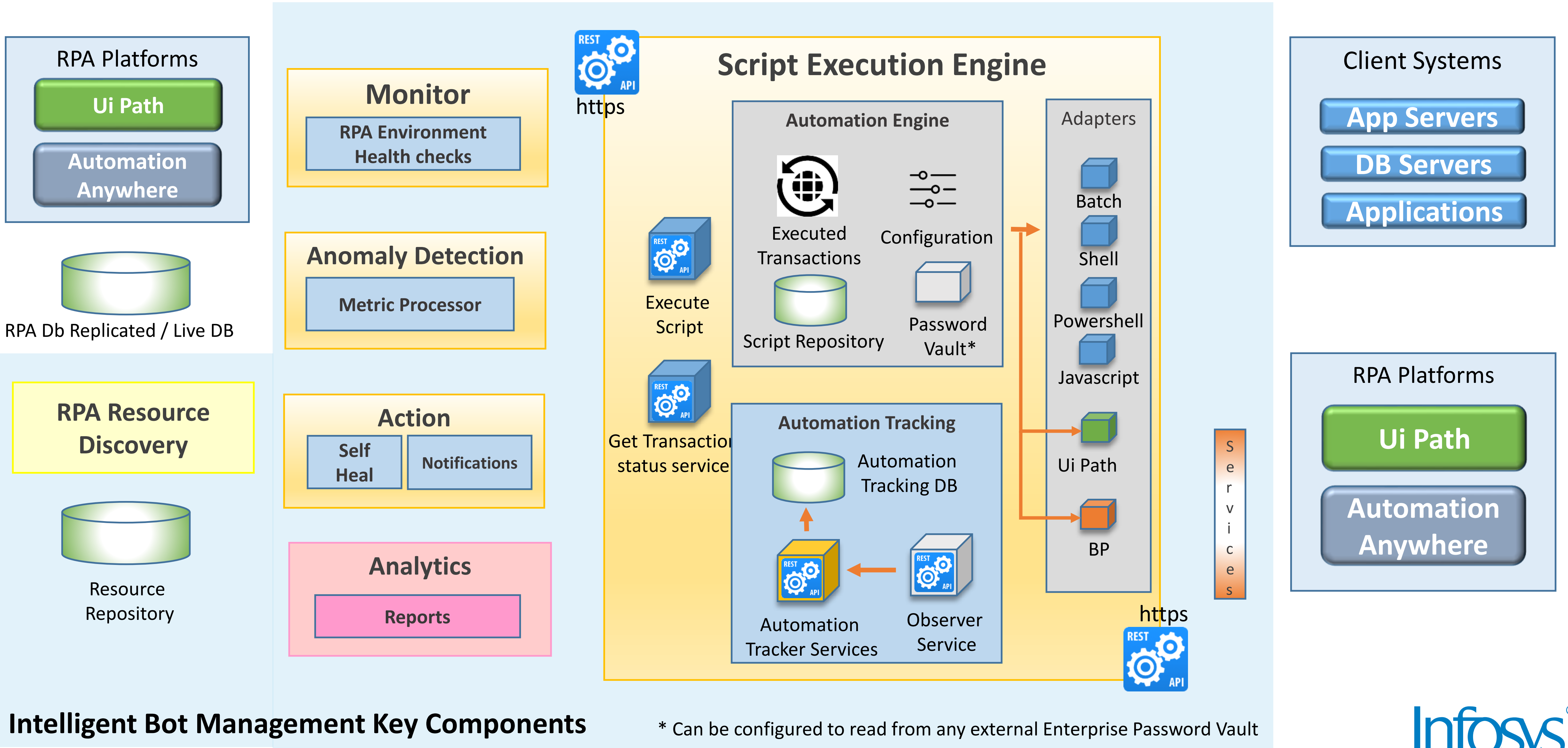
Infosys Intelligent Bot Management Solution Overview



* Automation Anywhere & UiPath Available.

Infosys Intelligent Bot Management High Level Architecture



* Can be configured to read from any external Enterprise Password Vault

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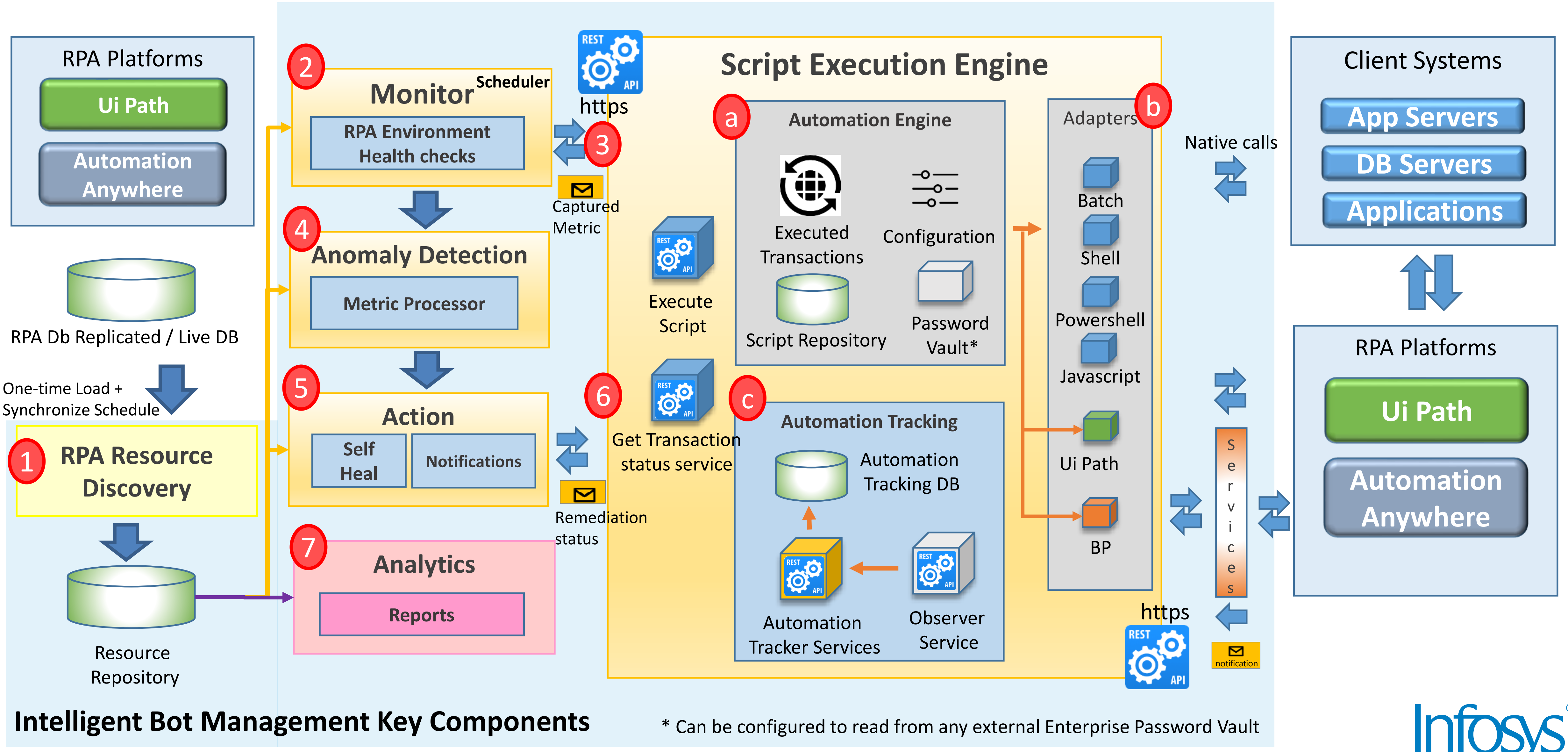
Infosys Intelligent Bot Management High Level Architecture



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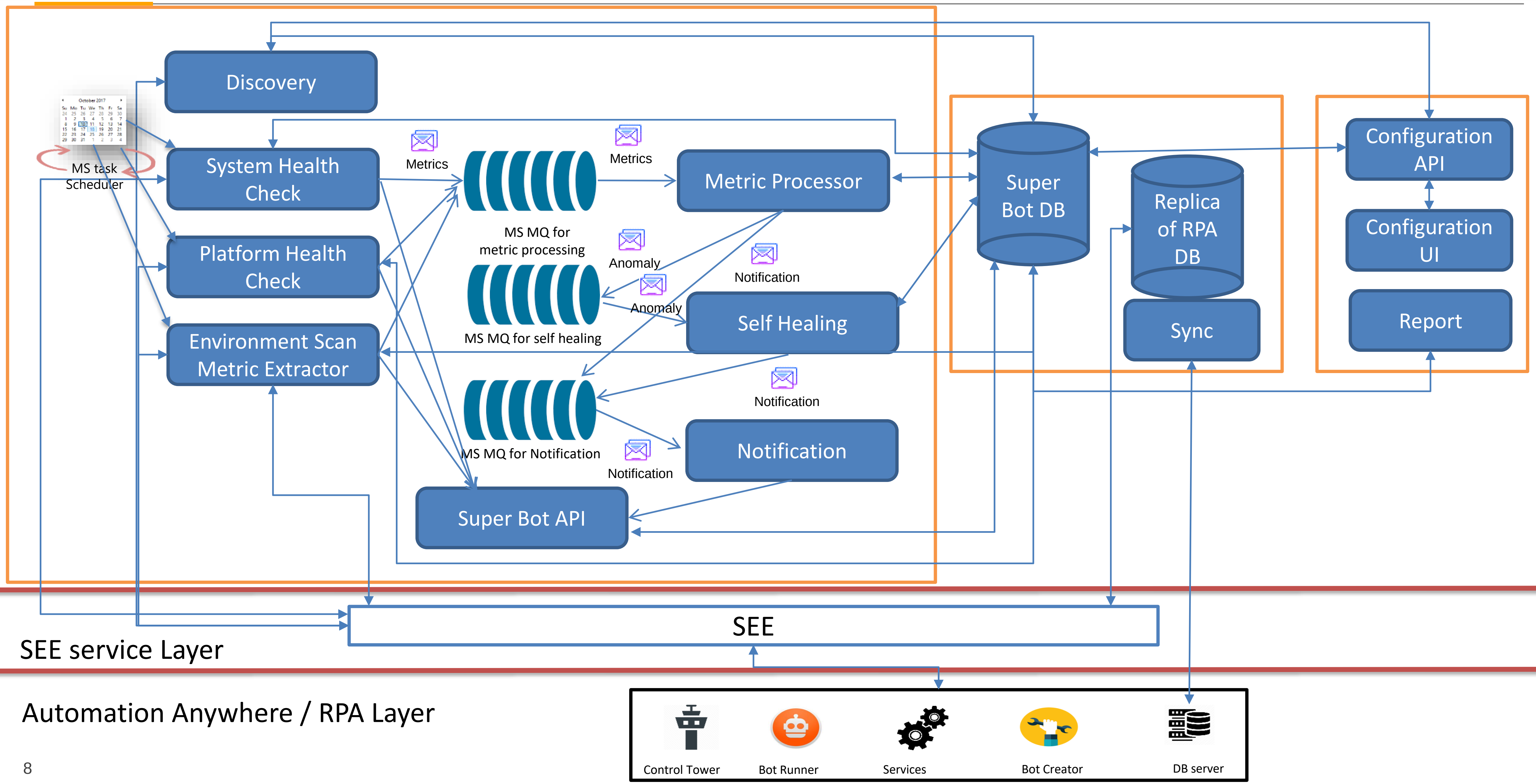
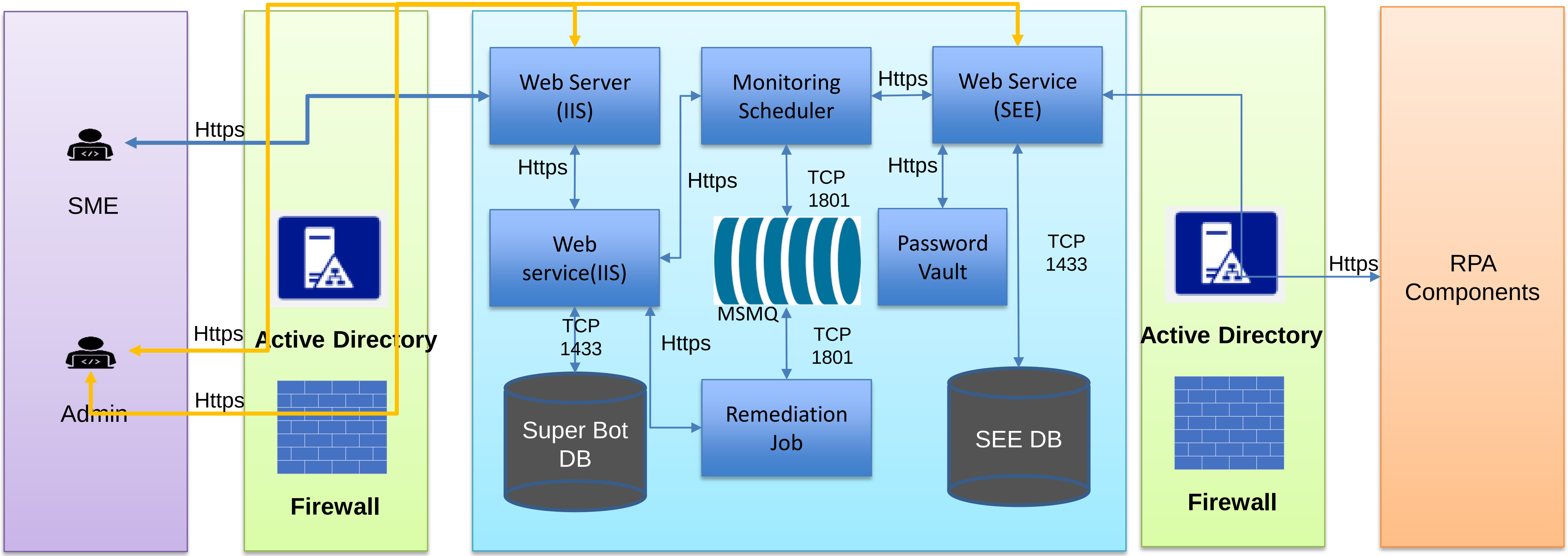
Infosys Intelligent Bot Management – Process Flow



SEE service Layer

Automation Anywhere / RPA Layer

Infosys Intelligent Bot Management Security Architecture



*HTTPS over default port 443

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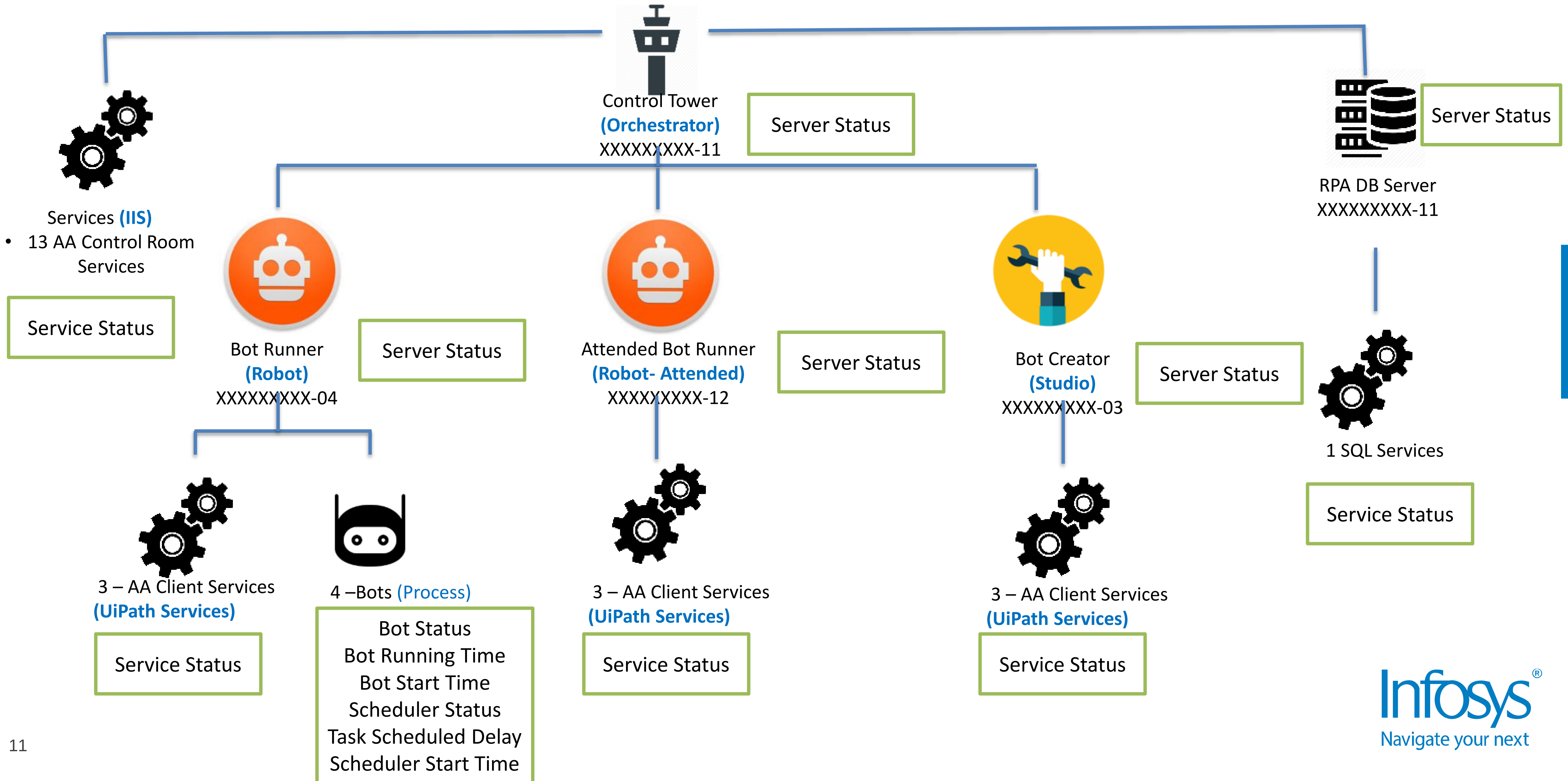
Demo 1 – Discovery and Configuration of Infosys Intelligent Bot Management

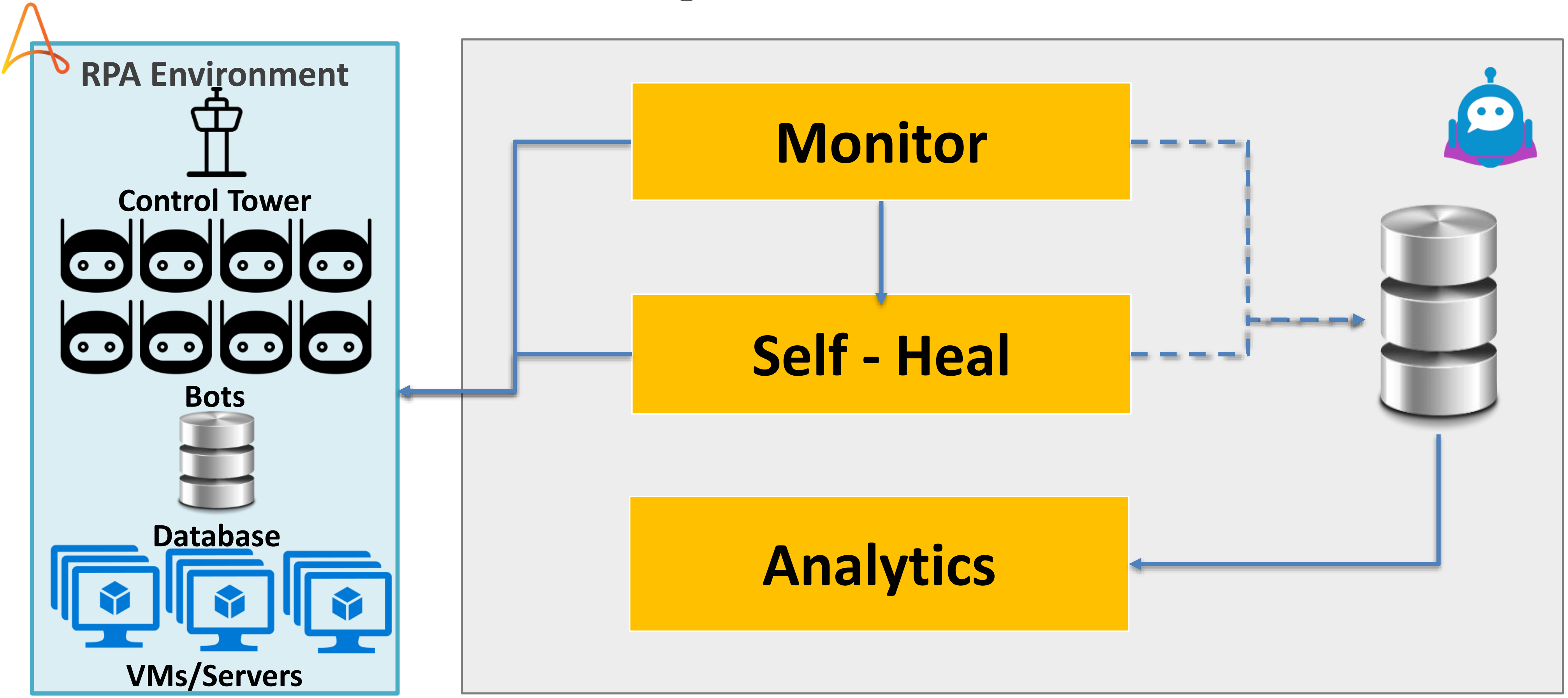
Infosys Intelligent Bot Management - Demo Model Configured



Demo 2 – Infosys Intelligent Bot Management operating in an RPA Environment

Scenario: Bot Outage Recovery Due to Failure of Dependent Component

Click to begin demonstration



Fault Re[illegible]gress!

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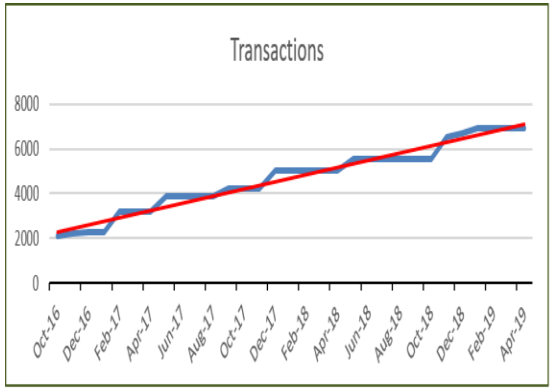
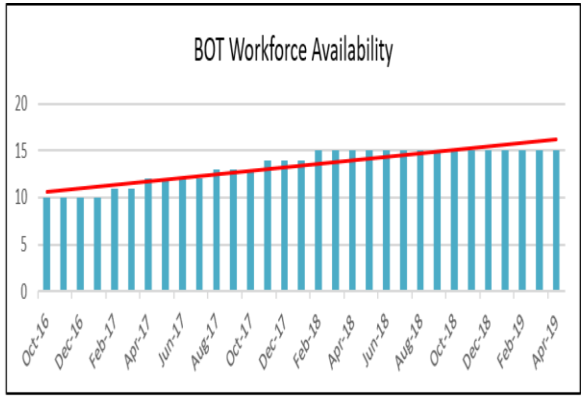
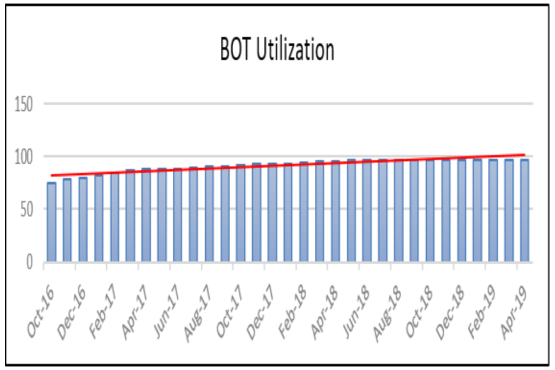
Demo 3 – Analytics Dashboards

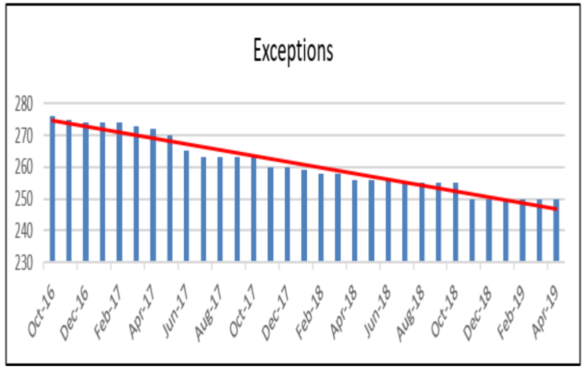
Analytics

Reports and Dashboards will be custom developed based on client specific requirements









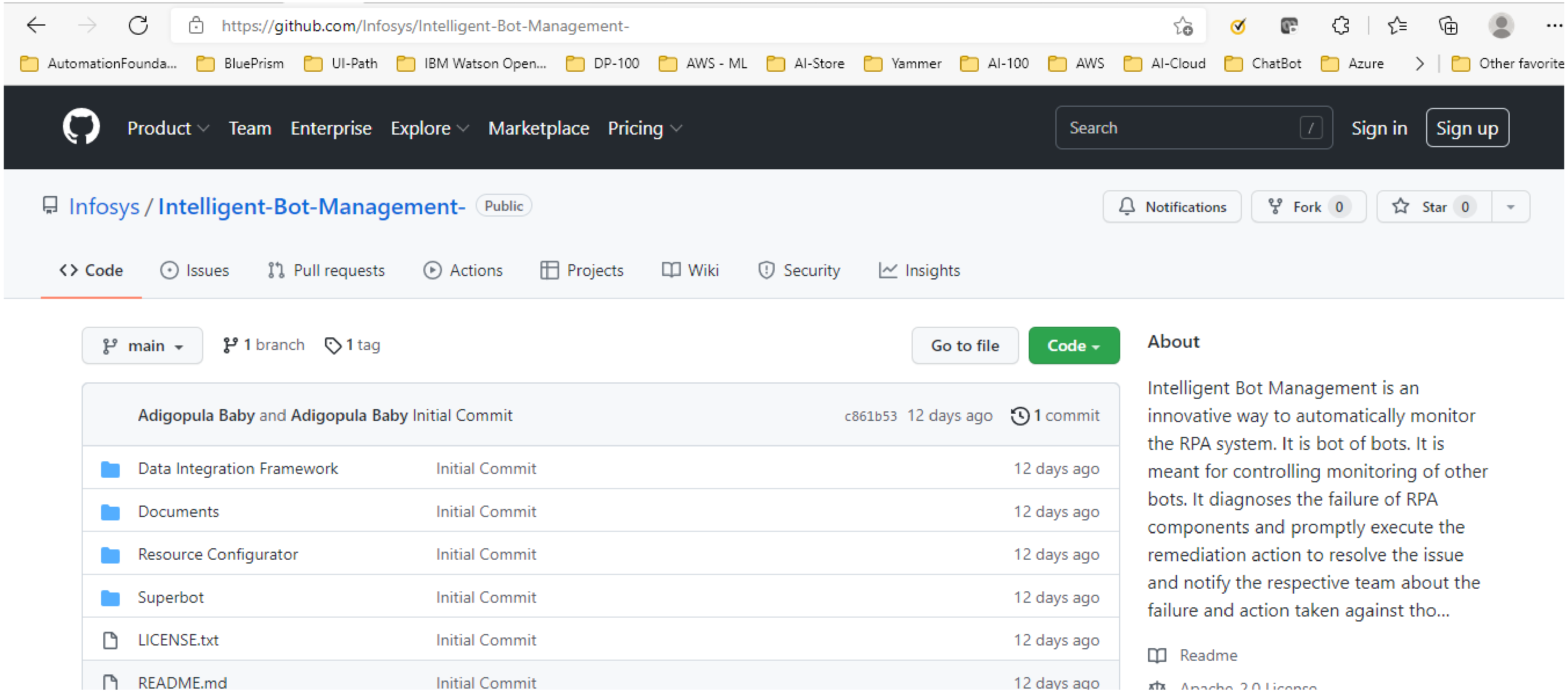
Opensource Support - GitHub

Solution is now available for public in Git Hub at following links

https://github.com/Infosys/Intelligent-Bot-Management-

https://github.com/Infosys/Script-Control-Center

https://github.com/Infosys/RPA-Control-Center



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Infosys Intelligent Bot Management – Hardware & Software Pre-requisites

Infosys Intelligent Bot Management – Minimum Hardware and Software Specification			
Server Type	**Quantity**	**Hardware configuration/ Server**	**Software Required/ Components**
Server	2	4 Core CPU, 8GB RAM, 500GB HDD	• Windows Server 2012 R2 – 64 Bit • SQL Server • MSMQ • MS Internet Information Services (IIS) • .NET Framework 4.6.1 • Windows Task Scheduler • Power BI Desktop • Powershell 3.0
Workstation – Developer	1	Dual Core CPU, 4GB RAM,HDD 500GB	• Windows 10 • .NET Framework 4.6 • Powershell 3.0

Pre-requisites:

1. Enterprise Active Directory enabled Service Account
2. Enterprise Exchange Server Access
3. RPA database and RPA API Read only access
4. Remote access to the clients and control room to read the service status
5. Access to run scripts remotely on RPA system nodes

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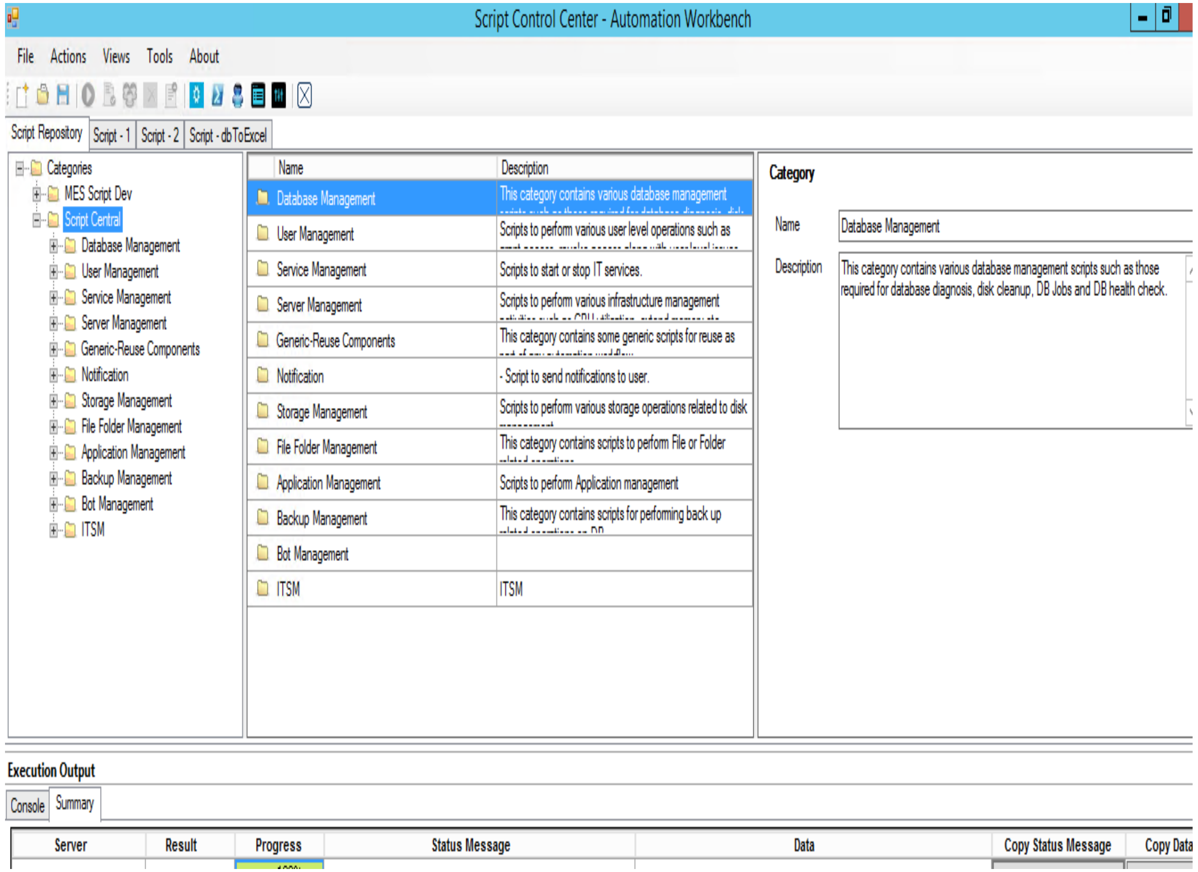
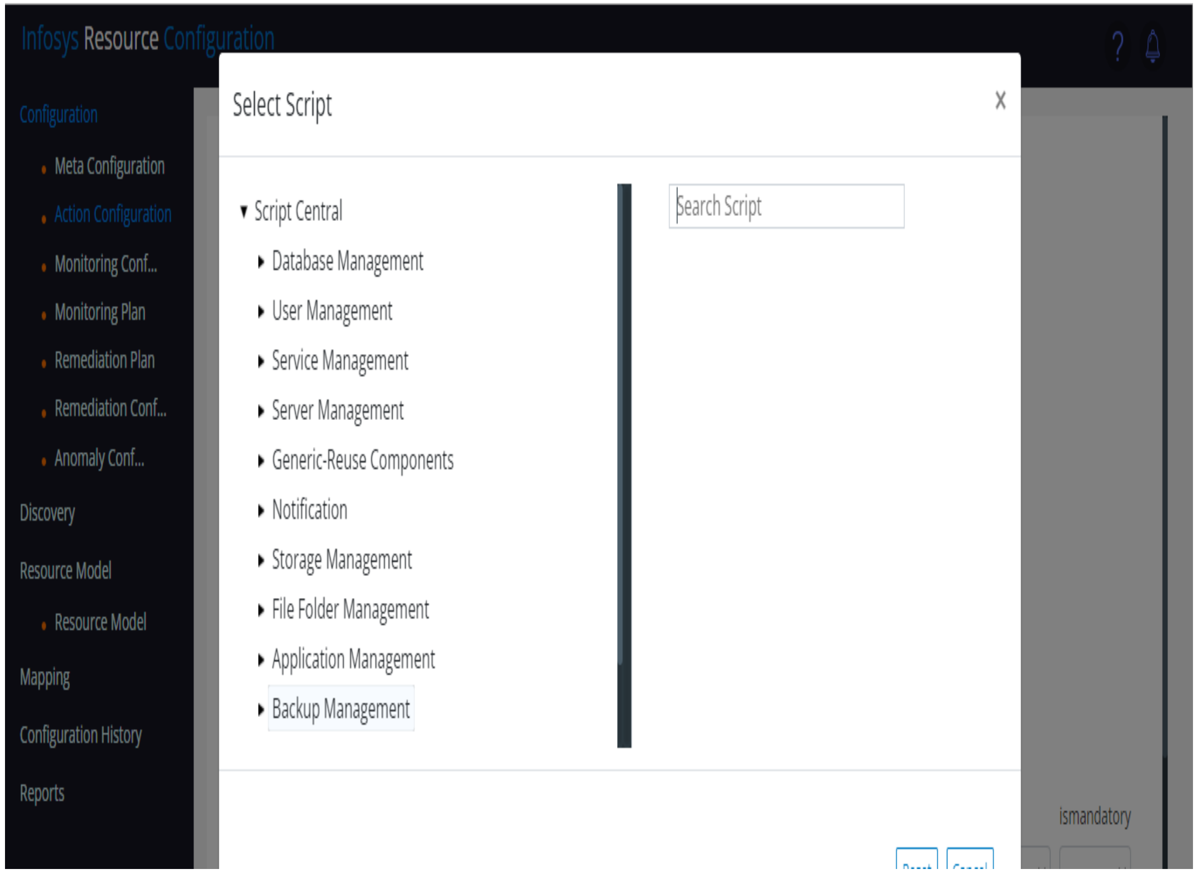
Appendix

Script Execution Engine - Pre-Configured Setup

- Script Repository is the central store to persist and manage all back-end execution scripts published by the end user, following structure with the respective pre-defined scripts are configured as past of Infosys Intelligent Bot Management

- Script Execution Engine execute back-end scripts maintained in repository in a controlled manner





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Pre-Configured Monitoring and Self-Healing setup for Automation Anywhere

Control Tower

- Server Availability
- Service Availability:
  - Control Room Service
  - Elastic Search Service
  - Control Room Reverse Proxy
  - Control Room Messaging
  - Control Room Caching
  - Bot Insight Visualization
  - Bot Insight Service Discovery
  - Bot Insight Service
  - Bot Insight Scheduler
  - Bot Insight Query Engine
  - Bot Insight Elastic Search
  - Bot Insight EDC
  - Licensing

Bot Runner n

- Server Availability
- Service Availability:
  - AAE_AutoLoginService_v11
  - AAE_SchedulerService_v11
  - AAE_ClientService_v11

Bot n

- Bot Status
- Bot Running Time
- Bot Start Time
- Scheduler Status
- Task Scheduled Delay
- Scheduler Start Time

Bot Creator n

- Server Availability
- Service Availability:
  - AAE_AutoLoginService_v11
  - AAE_SchedulerService_v11
  - AAE_ClientService_v11

Platform Database

- Server Availability
- Service Availability
  - MSSQLSERVER Service

* Automation Anywhere & UiPath Available

Pre-Configured Monitoring and Self-Healing setup for UiPath

Orchestrator

Metric
- Server Availability
- Service Status

Monitoring scripts
- Check Orchestrator Status
- Check Service Status

Remediation scripts
- Server Up Status Check
- Start Service

Robot VM "n"

Metric
- Server Status
- Service Status

Monitoring scripts
- Check Robot Status
- Check Service Status

Remediation scripts
- Server Up Status Check
- Start Service

Process " n "

Metric
- Process Status
- Process Running Time

Monitoring scripts
- Check Job Status
- Check Job Execution Time

Remediation scripts
- Start Job
- Manual Remediation

Platform Database

Metric
- Server Status
- Service Status

Monitoring scripts
- Check DB Server Status
- Check Service Status

Remediation scripts
- Server Up Status Check
- Start Service

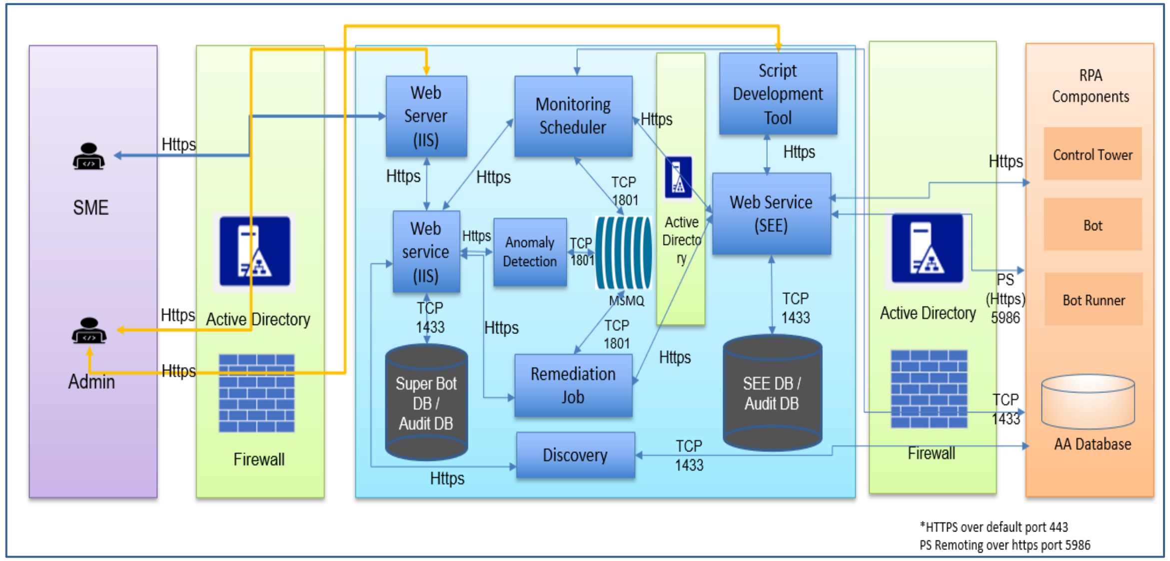
* Automation Anywhere & UiPath Available

Security Considerations – Access security within

Access to Infosys Intelligent Bot Management Components

- User logs in using the username and password using Windows based Authentication
- Upon successful authentication, the user request is allowed to proceed from Web Tier to Application Tier using the secure HTTPS Protocol.
- All access to the Intelligent Bot Management Database server is enabled via the web services alone.
- Script Execution Password details are stored in Encrypted format either in Data base/ Enterprise Password Vault
- Script Execution Engine interact with all RPA Components using authorized service account over Https and a firewall restricts access to only the required ports
- The automation scripts are protected from accidental execution. The scripts can only be run by the authorized service account.
- Role based access controls are supported, which can be configured to securely manage scripts published to the repository

Network Security



- Any interaction with the Intelligent Bot Management components is over HTTPS and a firewall restricts access to only the required ports of the systems, applications, servers, etc. that are involved in automation.
- All the User/System actions are logged for audit purpose

THANK YOU

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